

ADAPTIVE CONTINUOUS AUTHENTICATION (ACA) — MASTER GANTT SCHEDULE

UCS503P: Software Engineering Project • Thapar Institute of Engineering & Technology, Patiala • Supervisor: Dr. Jeelani Asif

Total Tasks

23

Completed

100%

Timeline

51 Days

Status

On Track

Task ID	Phase & Task Description	Lead	Dur.	Dates	Status	Aug 01 (W1)	Aug 08 (W2)	Aug 15 (W3)	Aug 22 (W4)	Aug 29 (W5)	Sep 05 (W6)	Sep 12 (W7)	Sep 19 (W8)
Phase 1: Problem Formulation & Threat Modeling		Schedule: Aug 01 - Aug 16, 2026											
T01	Literature Review on Keystroke Biometrics	Yatharth K.	5d	08/01-08/05	100%	5d							
T02	Threat Modeling: Workstation Hijacking & Token Theft	Yatharth K.	4d	08/06-08/09	100%		4d						
T03	Project Criteria Justification & Scope Definition	Anjali K.	3d	08/10-08/12	100%		3d						
T04	Formal LaTeX Proposal Document & Compilation	Ridhi B.	4d	08/13-08/16	100%			4d	★ M1: Proposal Submission				
Phase 2: Hardware & DOM Keystroke Collection Engine		Schedule: Aug 17 - Aug 31, 2026											
T05	OS-Level Global Keystroke Listener (pynput)	Yatharth K.	5d	08/17-08/21	100%			5d					
T06	Client-Side Ephemeral HMAC-SHA256 Anonymizer	Yatharth K.	3d	08/22-08/24	100%			3d					
T07	Initial User Baseline Capture (U01/S01, S02)	Yatharth K.	4d	08/25-08/28	100%				4d				
T08	Path Traversal & Security Validation Protocols	Ridhi B.	3d	08/29-08/31	100%					3d			
Phase 3: Biometric Feature Engineering & ML Pipeline		Schedule: Sep 01 - Sep 11, 2026											
T09	Hold (Dwell) and Flight Timing Calculation Algorithms	Yatharth K.	3d	09/01-09/03	100%					3d			
T10	10-Second Sliding Window Feature Aggregator	Yatharth K.	3d	09/04-09/06	100%						3d		
T11	Inactive Window Gating Policy (Min. 5 Keystrokes)	Anjali K.	2d	09/07-09/08	100%							2d	
T12	Scikit-Learn Preprocessing Pipeline (Imputer + Scaler)	Yatharth K.	2d	09/08-09/09	100%							2d	
T13	Isolation Forest Anomaly Model Training & Serialization	Yatharth K.	2d	09/09-09/10	100%								2d
T14	Continuous Linear Trust-Mapping Function (0-100%)	Yatharth K.	2d	09/10-09/11	100%							2d	★ M2: ML Pipeline & Models
Phase 4: Web Dashboard & Real-time Decision Integration		Schedule: Sep 11 - Sep 16, 2026											
T15	Flask REST API Gateway & Authentication Routes	Yatharth K.	2d	09/11-09/12	100%							2d	
T16	Responsive Security Cockpit Dark-Mode UI	Anjali K.	2d	09/12-09/13	100%								2d
T17	Dynamic SVG Radial Gauge & Live Historical Chart	Anjali K.	2d	09/13-09/14	100%								2d
T18	Dynamic 3-Tier Policy State Machine (ALLOW/VERIFY/LOCK)	Yatharth K.	2d	09/14-09/15	100%							2d	★ M3: Security Cockpit Live
T19	Replay Mode Controller for Ground-Truth Demos	Yatharth K.	2d	09/15-09/16	100%								2d
Phase 5: Automated Testing, Benchmarking & Deliverables		Schedule: Sep 16 - Sep 20, 2026											
T20	Automated Integration Test Suite (11 Tests, 100% Pass)	Ridhi B.	2d	09/16-09/17	100%								2d
T21	Latency & Performance Engineering Benchmarks (<1ms)	Yatharth K.	2d	09/17-09/18	100%								2d
T22	Mid-Semester Prototype LaTeX Report & Vector Diagrams	Ridhi B.	2d	09/18-09/19	100%								2d
T23	Presentation Slide Deck & Live Demonstration Rehearsal	Anjali K.	2d	09/19-09/20	100%								2d

Phase Color Codes: Phase 1 (Problem Formulation)

Phase 2 (Hardware)

Phase 3 (Biometric Feature Engineering)

Phase 4 (Web Dashboard)

Phase 5 (Automated Testing, Benchmarking)

Course Milestones: ★ Milestone 1 (Proposal) • ★ Milestone 2 (ML Engine) • ★ Milestone 3 (Decision Cockpit) • ★ Milestone 4 (Prototype Stage)

Workload Distribution: Yatharth Kansal (14 Tasks) • Anjali Kumari (5 Tasks) • Ridhi Batra (4 Tasks)